

Unit	Topic
I	Introduction to Forces: Introduction to Engineering Mechanics covering, Force Systems, Idealizations in Mechanics, Rigid Body Equilibrium; System of Forces, Coplanar Concurrent Forces, Components of Forces, Resultant, Moment of Forces and its Application; Couples and Resultant of Force System, Equilibrium of System of Forces, Free body diagrams, Equations of Equilibrium of Coplanar Concurrent and Non-Concurrent Force Systems. Simple Stress and Strain, Axially loaded members and Hooke's law.
II	Centroid: Introduction, Center of Gravity and Centroid, Centroid of Simple Figures from First Principle, Centroid of Composite Sections; Centre of Gravity and its Implications, Moments of inertia: Area and Mass Moment of Inertia, Product of inertia, Principal axes and Principal moments of inertia, Moment of Inertia of Plane Sections from First Principle, Theorems of Moment of Inertia, Moment of Inertia of Standard Sections and Composite Sections. Transformation of Moment of Inertia.
III	Structures Introduction, Classification, Analysis of Plane Trusses- Method of Joints, Method of Sections, Method of Tension Coefficients, Graphical Method, Beams-Its type, Shear force and Bending Moment Diagrams.
IV	Friction Introduction, Laws of Coulomb friction, Angle of friction, Angle of Repose, Cone of Friction, Sliding and Rolling Friction, Rope and Belt Friction, Screw Friction, Wedge Friction. Principle of Virtual work and Energy Strain Energy, Virtual Displacement, Principle of Virtual work, Mechanical Efficiency, Work of a force/couple (springs etc.), Potential Energy and equilibrium, stability.
V	Kinematics and Kinetics of Rigid Bodies Introduction, Types of motions in plane and space, Rotation of rigid bodies, General Plane motion, D'Alembert's Principle, Force, Mass and Acceleration, Work and Energy, Impulse and Momentum, Introduction to Free & Forced Vibrations.

Text/Reference Books:

1. Beer F.P. and Johnston E.R., Mechanics for Engineers-Volume I -Statics, Volume-II Dynamics, McGraw Hill, New York.
2. Merriam J.L and Kraige L.G., Engineering Mechanics, Volume I-Statics, Volume-II Dynamics, John Wiley & Sons, New York.
3. Shames L.H. Engineering Mechanics, Prentice Hall, New Delhi.
4. R. C. Hibbler Engineering Mechanics, Vol I and II Pearson Press 2002